Flash Flood Early Warning, Rescuer Assistance, and Safe Evacuation Routing System

A Practical Engineering Proposal for Sensor Failover, Street-Level Inundation Estimation, and Depth-Aware Evacuation Navigation

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Target Study Area:	Brahmaputra & Barak River Basin (Focusing on Kamrup Metropolitan & Flood-Prone Districts, Assam)

1. Introduction & Project Objective

During the monsoon season, sudden intense rainfall in the northeastern hill tracts of India causes rivers and surrounding tributaries to surge rapidly. In river basin regions like Assam, flash floods raise water levels over short periods, creating severe challenges for communities living near riverbanks and for municipal disaster agencies. The objective of this project is to develop an automated, practical software system that predicts street-level flood depths in advance, assists emergency rescue personnel (NDRF and SDRF) in positioning assets before access corridors submerge, and guides citizens along safe, dry routes to designated government relief shelters.

Rather than focusing on theoretical hydrodynamics alone, this project tackles the direct operational issues that cause disaster response to fail on the ground: loss of river gauge data during severe weather, general weather advisories that lack street-level depth details, standard GPS map apps that guide drivers into flooded roads, and reactive rescue deployments that arrive after road access is already cut off.

2. The Operational Issues We Are Fixing

Issue 1: Physical River Gauges Fail During Severe Storms

Automated river staff gauges and ultrasonic level sensors installed on riverbanks are exposed to strong currents, uprooted trees, floating logs, and heavy silt. These forces frequently damage or submerge the sensors. At the same time, high winds cause local power blackouts and cellular tower failures. As a result, ground sensor telemetry frequently stops transmitting during extreme storms. When physical sensors go offline, disaster managers lose real-time visibility into river levels at the exact time when water is rising fastest.

Issue 2: Standard Navigation Apps Lack Flood Depth Information

Commercial navigation apps (such as Google Maps) are designed to find the shortest or fastest dry-weather route based on traffic speed. They have no information about road water depth. During a flood, these apps continue directing fleeing families along roads that cross low underpasses, culverts, and river dips. Moving water as shallow as 15 cm can wash away pedestrians, while 30 cm floats ordinary passenger cars and stalls engines, creating serious hazards for evacuees.

Issue 3: Emergency Rescuers Are Dispatched After Access Roads Are Already Submerged

Emergency rescue teams are usually dispatched only after distress calls begin arriving. Many low-lying villages connect to regional highways via a single low-lying road. Once that access road is submerged under deep water, heavy rescue vehicles cannot enter, and launching motorized rescue boats through flooded trees and debris becomes slow and difficult. Rescue teams need advance notification to move boats and supplies into position before the access road is cut off.

3. System Architecture & Working Mechanism

The proposed system connects river sensor monitoring, automated satellite radar backup, road water depth calculation, and depth-aware navigation. The system operates across four coordinated stages:

Stage	Core Function & Implementation	Operational Outcome
1. River Monitoring & Satellite Radar Backup	Polls Central Water Commission (CWC) automated river gauges. Automatically switches to Copernicus Sentinel-3 SAR satellite radar altimetry if ground telemetry stops.	Maintains water height tracking through dense monsoon rain clouds without relying on vulnerable ground poles.
2. Downstream River Rise Estimation	Calculates flood wave travel along river reaches from upstream stations to downstream settlements; checks satellite soil moisture to identify when saturated soil creates direct runoff.	Provides roughly 2 or more hours of advance notice before the flood crest reaches downstream settlements.
3. Street-by-Street Flood Depth Mapping	Subtracts road elevation (from SRTM 30m Digital Elevation Model) from predicted river surface elevation across OpenStreetMap road centerlines.	Determines estimated water depth on every road segment and identifies single-access roads at risk of cut-off.
4. Multi-Channel Alerts & Depth-Aware Routing	Dispatches SMS alerts in local languages; routes evacuees along dry roads to verified government shelters; generates advance staging alerts for rescue teams.	Prevents families from entering flooded roads and enables rescue personnel to pre-stage boats before roads close.

3.1 Satellite Radar Backup: Maintaining Data When Ground Sensors Fail

Under normal conditions, the pipeline queries automated water level feeds from CWC river stations. The software performs continuous heartbeat checks: if telemetry from a ground gauge is missing for 15 minutes or reports corrupt values, the system flags the station as offline and activates backup satellite radar data.

Working Principle of Satellite Radar: Standard optical satellite cameras cannot see through dense monsoon clouds and rain. In contrast, Synthetic Aperture Radar (SAR) altimetry on satellites like Sentinel-3 sends microwave radar pulses that pass directly through thick storm clouds and heavy downpours. The radar beam reflects off the water surface back to the satellite, allowing water surface elevation to be measured from orbit. Because the satellite is in space, it is completely unaffected by damaged ground poles, local power cuts, or fallen cellular towers.

3.2 Downstream River Rise Estimation

When heavy rainfall occurs in upper hill catchments, the surge moves downstream along the river channel. By tracking water elevation at upstream stations and estimating flood wave travel time along the river reach, the system estimates the expected river height and arrival time at downstream towns, providing a lead time of 2 or more hours.

Soil Moisture Factor: Dry soil absorbs heavy rain, but after consecutive days of monsoon showers, soil becomes saturated. When soil moisture reaches saturation, additional rainfall cannot seep into the ground and turns immediately into rapid surface runoff. The system reads satellite soil moisture data (NASA GPM feeds) to identify when incoming rainfall will directly produce sharp surface surges.

3.3 Street-Level Flood Depth Calculation

To calculate water depth on individual roads, the system compares the predicted river water level against the ground elevation of each road segment using a 30-meter Digital Elevation Model (SRTM DEM) and OpenStreetMap road vector centerlines. The estimated water depth determines road accessibility:

Depth Category	Estimated Depth	Vehicle & Pedestrian Accessibility	System Action
Tier 1: Dry	< 5 cm	All civilian vehicles and pedestrians permitted.	Designated as open evacuation corridor.
Tier 2: Shallow Water	5 cm to 15 cm	Vehicles passable with care; unsafe for walking pedestrians.	Excluded from walking evacuation routes.
Tier 3: Dangerous Depth	15 cm to 80 cm	Closed for civilian cars and walkers; high rescue trucks only.	Reserved for NDRF/SDRF 4x4 rescue trucks.
Tier 4: Deep Flood	> 80 cm	Closed to all motor vehicles; rescue boats only.	Marked impassable for all wheeled traffic.

3.4 Emergency Rescuer Assistance: Cut-Off Alerts & High-Clearance Routes

A practical contribution of this project is assisting rescue agencies (such as NDRF and SDRF) before access is lost:

Advance Single-Access Cut-Off Alerts: Many riverside villages connect to main highways through a single road. The system identifies villages whose sole connecting road is predicted to exceed 30 cm depth within the next 2 hours. Disaster coordinators receive an automated alert so they can dispatch boats, fuel, and medical personnel to high-ground staging areas near those villages before the road is submerged.

High-Clearance Rescue Truck Routes: Heavy emergency trucks (NDRF 4x4 vehicles) have high air intakes and can safely ford water up to 80 cm deep. The routing engine calculates dedicated paths allowing rescue convoys to traverse shallow flooded streets to reach stranded residents directly.

3.5 Civilian Evacuation: Depth-Aware Safe Routing

The evacuation routing engine builds on OpenStreetMap road graphs using Dijkstra's algorithm with depth safety constraints:

- **Pedestrian Ceiling (< 15 cm):** Moving water as shallow as 15 cm can sweep away walking adults and children. For pedestrians, roads with 15 cm or more water are assigned infinite travel cost, guiding families strictly along dry ridges.
- **Vehicle Ceiling (< 30 cm):** For cars and two-wheelers, the system blocks road dips and underpasses where water exceeds 30 cm, preventing vehicle stall.
- **Dynamic Road Blocking:** As rainfall continues and water levels rise, flooded segments are dynamically closed on the graph, automatically rerouting evacuees along safe alternative roads.

3.6 Government Relief Centre Selection & Capacity Balancing

The system ensures families are guided only to shelters that are genuinely secure: (1) verifying that the shelter's plinth elevation is above the projected flood water level, (2) confirming that at least one incoming road connection remains dry (< 15 cm depth), and (3) monitoring shelter capacity so that once a relief center approaches its maximum occupancy, incoming evacuees are redirected to the next nearest open facility.

4. How This Idea Is New: Comparison with Existing Systems

The table below summarizes the key operational differences between existing disaster management tools and our proposed engineering solution:

Operational Dimension	Current Standard Systems	Proposed Solution (What Is New)
River Monitoring Continuity	Relies solely on physical river staff gauges; stops transmitting if sensors are damaged or storm power fails.	Automated failover to Copernicus Sentinel-3 SAR satellite radar altimetry; penetrates rain clouds without ground sensors.
Warning Granularity	Broad district-level or river basin warnings; does not specify which local streets will flood.	Calculates estimated water depth street-by-street by intersecting DEM elevation with predicted water surface.
Evacuation Navigation	Commercial GPS apps (Google Maps) only consider dry traffic; frequently route cars into submerged roads and underpasses.	Depth-constrained graph routing enforcing physical safety limits (<15 cm walking, <30 cm car) to avoid flooded roads.
Rescuer Support	Reactive deployment after distress calls; rescue teams often find access roads already submerged.	Flags single-access villages facing road cut-off 2 hours in advance; provides special routes for 4x4 rescue trucks (≤80 cm).
Relief Shelter Allocation	Static list of shelters without verifying road access or live capacity; evacuees may find facilities flooded or full.	Dynamically verifies dry road access to shelters and redirects evacuees when a facility approaches full capacity.

5. Implementation Plan & Open-Source Technology Stack

The project is built entirely with open-source software libraries and publicly available government and satellite datasets, ensuring the framework can be deployed by district disaster agencies without licensing costs:

Subsystem	Technologies & Datasets	Purpose in Project
Sensor Ingestion & Backup	Python, Pandas, Requests, Sentinel-3 SAR Altimetry API, NASA GPM IMERG rainfall feeds	Automates sensor polling and triggers satellite backup when ground feeds stop.
Hydrology & Inundation Core	NumPy, SciPy, SRTM 30m Digital Elevation Model (DEM), CWC River Reach cross-sections	Models downstream wave propagation and maps planar flood depth over terrain.
Road Network & Routing	OSMnx, NetworkX, OpenStreetMap Road Vectors, Depth-Constrained Dijkstra Graph Search	Ingests road segments and calculates depth-constrained safe routes.
Alerting & Web Interface	FastAPI, Leaflet.js, OpenStreetMap Tiles, OASIS CAP v1.2 XML generator, Twilio SMS API	Renders live flood map and dispatches regional SMS warnings in local languages.

6. Project Modules & Development Milestones

Phase & Timeline	Key Milestones & Deliverables	Assigned Module Lead
Phase 1: Ingestion & Backup (Months 1-2)	CWC river sensor API integration, Sentinel-3 SAR data pipeline, and 15-minute polling loop implementation.	**[Module 1 Lead / Investigator 1]** Sensor data ingestion & failover
Phase 2: Hydraulics & GIS (Months 3-4)	Downstream wave celerity calculation, SRTM 30m DEM processing, and road depth intersection algorithm.	**[Module 2 Lead / Investigator 2]** Hydrologic modeling & GIS depth mapping
Phase 3: Routing & Shelters (Months 5-6)	Depth-constrained Dijkstra pathfinding on OSM graph, pedestrian/rescue profile filters, and shelter load balancing.	**[Module 3 Lead / Investigator 3]** Graph routing & shelter capacity logic
Phase 4: Alerts & Testing (Months 7-8)	OASIS CAP v1.2 alert feed, multi-language SMS dispatch, web situational dashboard, and integrated system testing.	**[Entire Project Group]** Integrated testing & documentation

7. Expected Impact & Conclusion

By focusing on the practical challenges observed during floods in Assam, this proposed system offers a realistic solution for early warning and evacuation. Autonomous satellite radar backup prevents monitoring loss when ground sensors break; predicting downstream river rise gives families advance notice to evacuate; identifying vulnerable access roads allows rescue teams (NDRF) to stage boats before roads submerge; and depth-aware navigation prevents fleeing citizens from driving or walking into flooded roads.

The proposed system can serve as an affordable, scalable, and automated disaster management tool for district authorities.

Key References & Foundational Literature

1. Central Water Commission (CWC), Ministry of Jal Shakti, Government of India, *'Hydrodynamic Modeling and River Flood Forecasting Guidelines'*, 2021.

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4. OpenStreetMap Contributors & Boeing, G., *'OSMnx: New methods for acquiring, constructing, and analyzing complex street networks'*, Computers, Environment and Urban Systems, 2017.